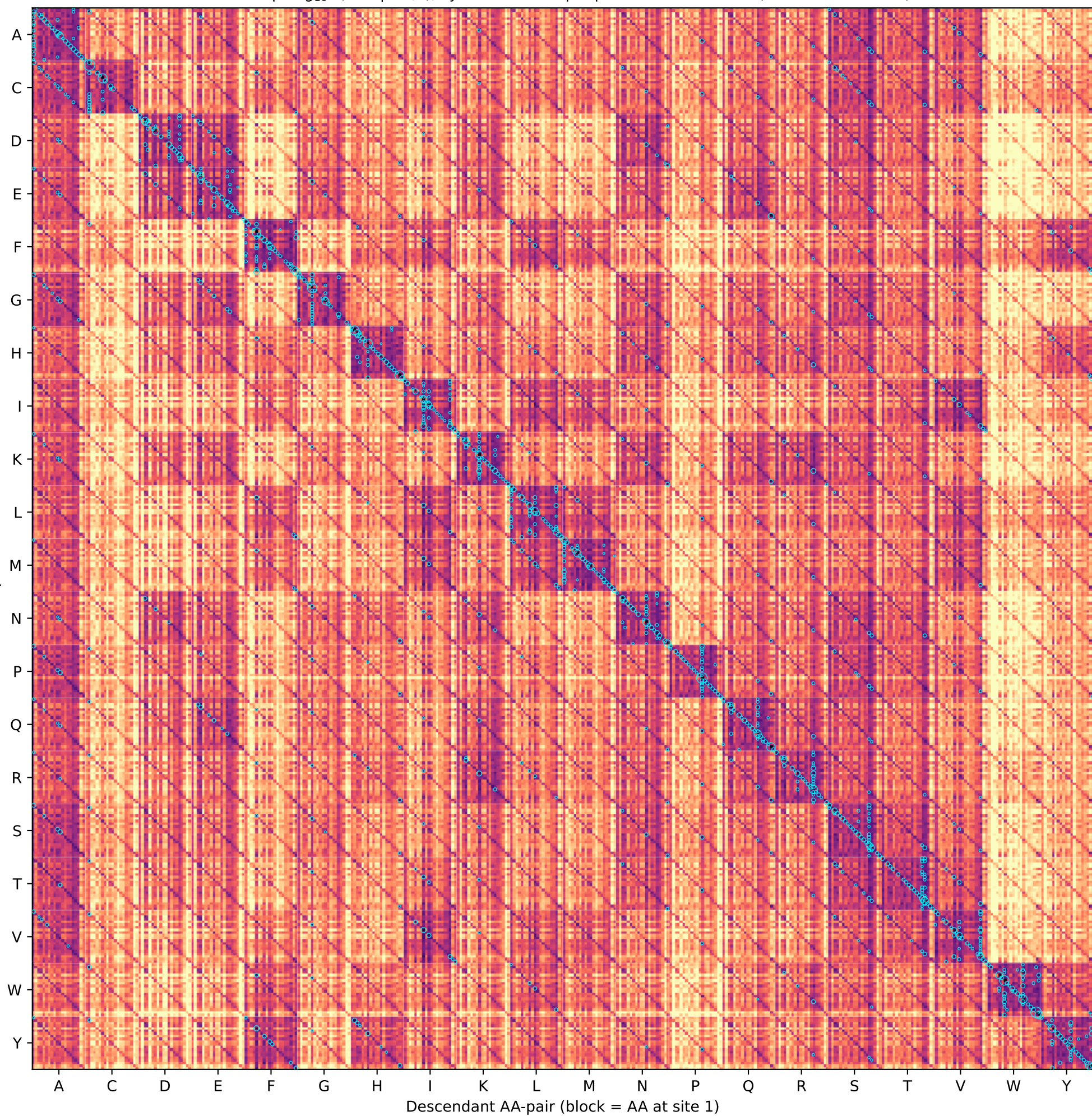


K=4 effective doublet substitution matrix at $t = 1.0$ (canonical: LG08 pair bg, empirical π_c)
 heatmap: $\log_{10} P(\text{desc}|\text{anc}; t)$; cyan bubbles: top-3 per row with $P \geq 0.02$ (total 1196 bubbles)

Ancestor AA-pair (block = AA at site 1)



$\log_{10} P(\text{desc. pair} | \text{anc. pair}; t)$

0
-1
-2
-3
-4
-5